Alternating Least Squares for Chebyshev CP Decomposition

1 Problem Setup

Consider the bivariate function

$$f(x, y) = \frac{1}{1 + c^2(x^2 + y^2)}, \quad (x, y) \in [-1, 1]^2,$$

with parameter $c > 0$. The goal is to approximate f on the Chebyshev grid using a low-rank Canonical Polyadic (CP) decomposition in terms of Chebyshev basis functions.

1.1 Chebyshev Nodes and Basis

Discretize $[-1, 1]$ using N Chebyshev nodes

$$x_j = \cos\left(\frac{\pi j}{N-1}\right), \quad j = 0, 1, \ldots, N-1,$$

and similarly for y. The corresponding Chebyshev Vandermonde matrices are

$$T_x = \big[T_k(x_j)\big]_{j,k}, \quad T_y = \big[T_\ell(y_m)\big]_{m,\ell},$$

where T_k denotes the kth Chebyshev polynomial.

1.2 CP Decomposition Ansatz

Approximate the discretized function $F \in \mathbb{R}^{N \times M}$ by

$$F \approx \sum_{r=1}^{R} \lambda_r \, (T_x a_r)(T_y b_r)^\top,$$

where $a_r \in \mathbb{R}^{d_x+1}$, $b_r \in \mathbb{R}^{d_y+1}$ are coefficient vectors in the Chebyshev basis, and $\lambda_r \in \mathbb{R}$ are scalar weights.

2 Alternating Least Squares (ALS)

The approximation is found by minimizing the mean squared error

$$\min_{\{a_r,b_r,\lambda_r\}} \left\| F - \sum_{r=1}^{R} \lambda_r (T_x a_r)(T_y b_r)^\top \right\|_F^2.$$

ALS proceeds by cyclically updating a_r, b_r, and λ_r while holding the others fixed.

2.1 Update of a_r

Fix $\{b_k, \lambda_k\}_{k\neq r}$. Define the residual

$$F_r = F - \sum_{k\neq r} \lambda_k (T_x a_k)(T_y b_k)^\top.$$

Then a_r is updated by solving the regularized least-squares problem

$$\min_{a_r} \left\| F_r - \lambda_r (T_x a_r)(T_y b_r)^\top \right\|_F^2 + \epsilon \|a_r\|_2^2,$$

leading to the normal equations

$$(T_x^\top T_x + \epsilon I) a_r = T_x^\top (F_r (T_y b_r)),$$

followed by normalization $\|a_r\|_2 = 1$.

2.2 Update of b_r

Analogously, the update for b_r is

$$(T_y^\top T_y + \epsilon I) b_r = T_y^\top (F_r^\top (T_x a_r)),$$

with normalization $\|b_r\|_2 = 1$.

2.3 Update of λ_r

Finally, the scalar λ_r is updated by projecting:

$$\lambda_r = \frac{\langle F, (T_x a_r)(T_y b_r)^\top \rangle}{\|(T_x a_r)(T_y b_r)^\top\|_F^2}.$$

To ensure numerical stability, λ_r is clipped within a prescribed range.

3 Termination and Evaluation

The process iterates until convergence or divergence is detected. The training error is measured via the Frobenius norm

$$\mathrm{RMSE}_{\mathrm{train}} = \sqrt{\frac{1}{NM}\|F - F_{\mathrm{hat}}\|_F^2},$$

and generalization is tested on a dense evaluation grid, to compute

$$\mathrm{RMSE}_{\mathrm{eval}} = \sqrt{\frac{1}{100^2}\sum_{i,j}\big(f(x_i, y_j) - f_{\mathrm{hat}}(x_i, y_j)\big)^2}.$$

In my code the function is

$$f(x, y; C) = \frac{1}{1 + \|C \begin{bmatrix} x \\ y \end{bmatrix}\|^2}.$$

- $C = \alpha I$ gives a symmetric circular dome, which we were initially using.
What initial C do you recommend I use?

$$C = D\,R(\theta), \quad D = \mathrm{diag}(a,b), \quad R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

- **Vertical (elongated vertically):**

$$C = \begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}$$

- **Horizontal (elongated horizontally):**

$$C = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}$$

- **Tilted (ellipse rotated by $\theta = 30^\circ$, with $a = 1, b = 5$):**

$$D = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}, \quad R(30^\circ) = \begin{bmatrix} 0.8660 & -0.5000 \\ 0.5000 & 0.8660 \end{bmatrix}$$

$$C = D\,R(30^\circ) = \begin{bmatrix} 0.8660 & -0.5000 \\ 2.5000 & 4.3301 \end{bmatrix}$$

- **Level curves very close (steeper dome overall):**

$$C = 5I = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

- **Level curves very far (flatter dome overall):**

$$C = 0.2I = \begin{bmatrix} 0.2 & 0 \\ 0 & 0.2 \end{bmatrix}$$

Results Summary

N = 64
Training carried out on Chebyshev nodes
Testing carried out on a uniform 100×100 grid

- $C = \begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}$ (vertical elongation):
  Train RMSE $= 6.0 \times 10^{-16}$, MaxE $= 4.9 \times 10^{-15}$;
  Train RelRMSE $= 2.4 \times 10^{-15}$, RelMaxE $= 4.9 \times 10^{-15}$;
  Test RMSE $= 5.8 \times 10^{-7}$, MaxE $= 2.4 \times 10^{-6}$;
  Test RelRMSE $= 1.8 \times 10^{-6}$, RelMaxE $= 2.4 \times 10^{-6}$.

- $C = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}$ (horizontal elongation):
  Train RMSE $= 4.8 \times 10^{-16}$, MaxE $= 4.1 \times 10^{-15}$;
  Train RelRMSE $= 1.9 \times 10^{-15}$, RelMaxE $= 4.1 \times 10^{-15}$;
  Test RMSE $= 5.8 \times 10^{-7}$, MaxE $= 2.4 \times 10^{-6}$;
  Test RelRMSE $= 1.8 \times 10^{-6}$, RelMaxE $= 2.4 \times 10^{-6}$.

- $C = \begin{bmatrix} 0.866 & -0.500 \\ 2.500 & 4.330 \end{bmatrix}$ (tilted ellipse, 30° rotation):
  Train RMSE $= 6.4 \times 10^{-16}$, MaxE $= 5.4 \times 10^{-15}$;
  Train RelRMSE $= 2.4 \times 10^{-15}$, RelMaxE $= 5.4 \times 10^{-15}$;
  Test RMSE $= 8.8 \times 10^{-8}$, MaxE $= 7.2 \times 10^{-7}$;
  Test RelRMSE $= 2.6 \times 10^{-7}$, RelMaxE $= 7.2 \times 10^{-7}$.

- $C = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$ (steeper dome):
  Train RMSE $= 2.8 \times 10^{-16}$, MaxE $= 2.4 \times 10^{-15}$;
  Train RelRMSE $= 2.4 \times 10^{-15}$, RelMaxE $= 2.4 \times 10^{-15}$;
  Test RMSE $= 3.7 \times 10^{-7}$, MaxE $= 2.3 \times 10^{-6}$;
  Test RelRMSE $= 2.1 \times 10^{-6}$, RelMaxE $= 2.4 \times 10^{-6}$.

- $C = \begin{bmatrix} 0.2 & 0 \\ 0 & 0.2 \end{bmatrix}$ (flatter dome):
  Train RMSE $= 9.5 \times 10^{-16}$, MaxE $= 3.2 \times 10^{-15}$;
  Train RelRMSE $= 9.8 \times 10^{-16}$, RelMaxE $= 3.2 \times 10^{-15}$;
  Test RMSE $= 1.1 \times 10^{-15}$, MaxE $= 5.2 \times 10^{-15}$;
  Test RelRMSE $= 1.1 \times 10^{-15}$, RelMaxE $= 5.2 \times 10^{-15}$.